

PRELIMINARY PROJECT PROPOSAL

Traffic Intensity and Road Maintenance Need in England, 2008–2025

1. Research Question and Basic Information

To what extent is road traffic intensity associated with the proportion of local A-road and local-authority-managed motorway length classified as red across English local authorities between FYE 2008 and 2025?

Unit of analysis. The local-authority-year. Each observation pairs one authority with one financial year ending (FYE).

Course	CISC7204 Data Analysis and Visualization	Region	England, United Kingdom
Team	Member 1: [ZHU QINYI / MC648900] Member 2: [XU XIAOYAN / MC647246]	Domain	Transportation – Road Infrastructure

2. Project Purpose

Why this topic matters. This project uses two official UK transport datasets to study the relationship between traffic intensity and road maintenance need across English local authorities. The project gives us a practical setting to apply the data analysis and visualization methods covered in CISC7204.

Outcomes. The project will produce a cleaned and integrated dataset for local authorities in England from 2008 to 2025. We will use this dataset to examine the direction and strength of the relationship between traffic intensity and road maintenance need. The project will connect the course methods with a real data problem. It will also give us experience in working with public-sector data that come from different sources and structures.

Skills. We will practise data cleaning, reshaping, feature engineering, and dataset merging with Python. We will use pandas and NumPy to prepare the data and create the traffic-intensity variable. We will use Matplotlib and Seaborn to explore distributions, time trends, regional differences, and relationships between variables. We will also use correlation and regression methods to examine the research question. We will compare model performance with training and testing data when appropriate.

Knowledge. The project will strengthen our understanding of the main stages of a data science workflow. We will learn how data types, missing values, variable definitions, and dataset structure can affect an analysis. We will also develop our understanding of exploratory data analysis, descriptive statistics, correlation, and regression. The project will help us choose suitable visualizations for different types of variables and analytical questions. We will also learn how to interpret patterns, outliers, fitted relationships, and model evaluation results. We will distinguish statistical association from causation when we interpret the findings.

3. Data Sources and Feasibility

Feasibility summary: Both sources are official Department for Transport datasets and contain the geographic and temporal identifiers needed for a joint analysis. The common analysis period is 2008–2025.

Dataset	Coverage / variables	Role in project	Official source
RDC0120 – Local-authority-managed classified road where maintenance should have been considered (red)	England; FYE 2008–2025. Main outcome: percentage of local A-road and local-authority motorway length categorised red.	Outcome variable, and comparison across regions and years.	RDC0120 ODS file – Road condition statistics: data tables (RDC), GOV.UK
Local authority traffic by vehicle class	Great Britain; 1993–2025. Local-authority code, year, road length, cars and taxis, and all motor vehicles (vehicle-miles).	Used to construct the traffic-intensity predictor and to align it with the road-condition observations.	Local authority traffic CSV – Road traffic statistics: bulk downloads, Department for Transport

4. Project Tasks and Methodology

- 1) **Data acquisition and audit:** The team will import the two official datasets and inspect their structure, variables, units, geographic coverage, and time periods. We will also identify missing-value conventions and relevant administrative notes.
- 2) **Road-condition cleaning and reshaping:** The team will use the RDC0120 A-roads and motorways worksheet. Values marked [x], [y], and [z] will be treated as unavailable or non-comparable rather than as zero. Aggregate rows will be removed, labels will be cleaned, and the financial-year columns will be reshaped from wide to long format.
- 3) **Traffic filtering and feature engineering:** The traffic dataset will be filtered to England and to the common period from 2008 to 2025. Traffic intensity will be calculated as annual motor-vehicle traffic divided by road-network length, giving annual vehicle-miles per road-mile.
- 4) **Integration:** The two datasets will be merged using the ONS local-authority code and year. The team will check code changes caused by local-authority reorganisation before merging records across time.
- 5) **Exploratory analysis:** The analysis will examine distributions, missing values, outliers, annual trends, and regional differences. Scatter plots, time-series plots, and regional comparisons will be used to identify patterns. Annual averages will be reported as unweighted means across available local authorities.
- 6) **Association and modelling:** Pearson correlation will be used as a descriptive measure of the relationship between traffic intensity and maintenance need. Regression models will then include traffic intensity together with year and region controls. Categorical variables will use a reference category to avoid redundant dummy variables.
- 7) **Evaluation and robustness:** The models will be evaluated using R^2 and MSE on held-out data. Cross-validation will provide an additional check of model stability. A temporal

holdout will also be considered so that earlier years are used for training and later years for testing.

- 8) **Interpretation and reporting.** The final analysis will report the direction and strength of the estimated association. The interpretation will distinguish association from causation and will discuss limitations such as road-class mismatch, missing observations, administrative changes, and unobserved factors.

Primary predictor	Traffic intensity = annual all-motor-vehicle traffic (vehicle-miles) ÷ road-network length (miles).
Primary outcome	Percentage of local A-road and local-authority motorway length categorised red, that is, length that should have been considered for maintenance.

5. Preliminary Visualisation and Justification

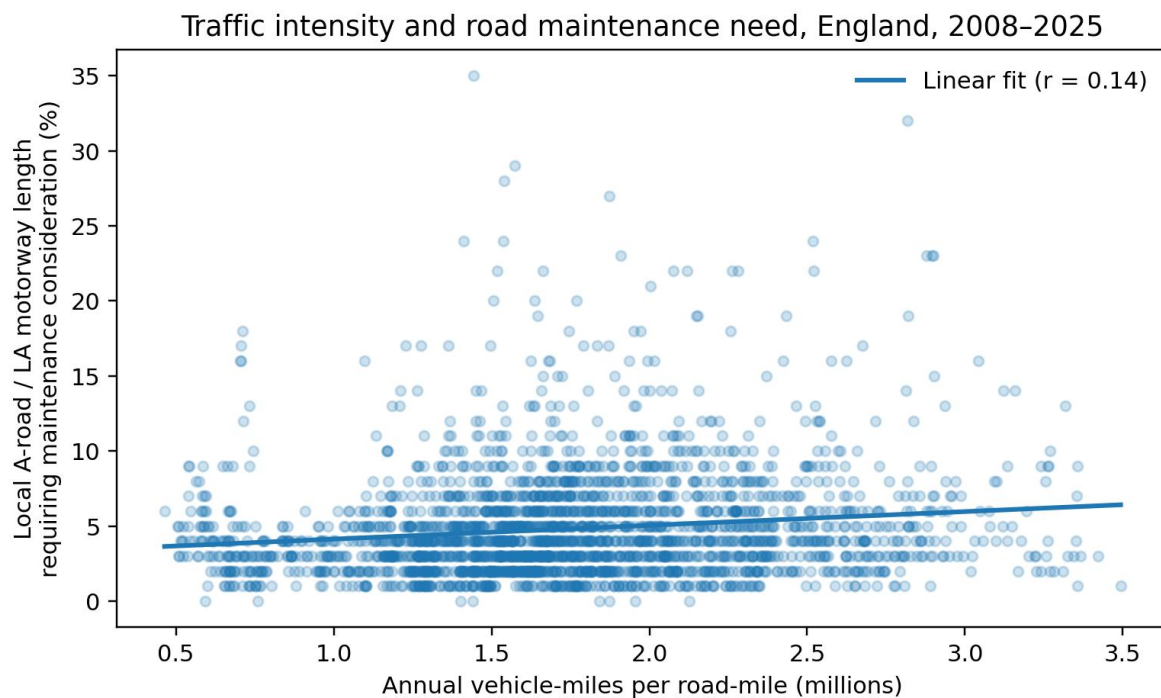


Figure 1. Preliminary pooled relationship between traffic intensity and maintenance need, England local-authority-year observations, 2008–2025.

Figure 1 compares traffic intensity with the percentage of local A-road and local-authority-managed motorway length classified as red. Each point represents one local authority in one year between 2008 and 2025. Traffic intensity is measured as annual vehicle-miles per road-mile, while the vertical axis represents the percentage of road length for which maintenance should have been considered. The fitted line is included to show the overall direction of the relationship without hiding the spread of individual observations.

The preliminary plot suggests that the pooled relationship is positive but weak ($r \approx 0.14$). The fitted line rises only slightly, and authorities with similar traffic intensity often have very different maintenance percentages. This means that traffic intensity may be related to road

condition, but it is unlikely to explain the variation on its own. The project will therefore examine whether the pattern remains after year and regional differences are taken into account. Because the data are observational, the analysis will describe association rather than claim that traffic causes road deterioration.

Dataset links. The two official datasets used in this project are available from the following sources. 1. [Road condition statistics: data tables \(RDC\), GOV.UK](#). 2. [Road traffic statistics: bulk downloads, Department for Transport](#). The analysis dataset derived from these two sources is exported by the notebook as `england_a_road_clean.csv`.

Verification notebook . The file `CISC7204_Traffic_Road_Condition_Project.ipynb` reproduces every figure and every statistic cited in this proposal, and it runs under Python 3.

6. Criteria for Success

- The project should successfully clean and integrate the two datasets at the local-authority-year level. The main data-processing decisions should be clearly documented.
- The project should define the traffic-intensity measure and road-condition outcome clearly and consistently.
- The analysis should use suitable statistical methods and visualizations to address the research question.
- The project should evaluate the reliability of the results with appropriate model evaluation methods and should consider the structure and limitations of the data.
- The final interpretation should be supported by the analysis. The project will be considered successful whether the relationship is strong, weak, or absent, as long as the findings are reported accurately and the main limitations are clearly explained.

Planned deliverables. The deliverables are a cleaned and merged CSV file, namely `england_a_road_clean.csv`; a reproducible Python notebook; a clearly labelled core visualisation with supporting charts; a concise summary of the statistical results and the model evaluation; and a final written interpretation that answers the research question and documents the limitations.

Criterion	Level 4 – Exemplary	Level 3 – Proficient	Level 2 or 1 – Developing
Research question and topic	Question is precise, measurable, and clearly scoped.	Question is clear and measurable with minor ambiguity.	The question is vague, too broad or not measurable.
Data cleaning and integration	Data are correctly cleaned, merged, documented, and reproducible.	Cleaning correct with minor undocumented choices.	Cleaning incomplete or not reproducible.
Exploratory analysis	Key patterns are examined with accurate and clearly labelled figures.	The exploratory analysis covers the main required views.	The exploratory analysis is superficial or misleading.
Model development and evaluation	Suitable models are compared using held-out data and cross-validation.	Main model evaluated on held-out data.	Evaluation is missing, leaks information, or is mis-specified.

Interpretation and limitations	The conclusion matches the evidence and clearly states uncertainty and limitations.	The conclusion broadly matches the evidence, with some limitations noted.	The conclusion overstates the evidence or omits the limitations.
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7. Collaboration Plan

Role	Primary responsibilities	Cross-review / shared responsibility
Member A – Data engineering lead	Import datasets; inspect data structure; clean missing values; reshape data; create variables; merge datasets by local authority and year.	Member B reviews the cleaning steps, variable definitions, and merged dataset.
Member B – Analysis and visualisation lead	Exploratory analysis; core and supporting figures; correlation and regression; model evaluation; interpretation and figure captions.	Member A reviews model inputs, figures, labels, and interpretation of results.
Shared	Refine the research question; discuss limitations; interpret findings; prepare the proposal and final submission.	Both members check the notebook, figures, conclusions, and final document before submission.

Ground rules. Both members will keep the raw data unchanged and use shared files for all project work. Each member will review the other member's work before a section is completed. The team will record important analytical decisions in the notebook. Both members will discuss any disagreement by referring to the research question and the original data definitions.

Team structure. Member A will act as the project coordinator and manage the project schedule. Member B will act as the project scribe and maintain the notebook records. Both members will share the progress tracker role. Member B will act as the project liaison for communication with the teaching team.

Use of Generative AI

We acknowledge the use of **ChatGPT (OpenAI)** to support background research and independent study, to generate materials that were reviewed and adapted for this proposal, and to assist with copy-editing and proofreading. ChatGPT was also used to clarify data science concepts, support Python debugging, and discuss data-cleaning and visualization methods. The team independently conducted and verified the data processing, analysis, visualizations, and model evaluation. All AI-assisted material was critically reviewed and revised before inclusion. The team takes full responsibility for the accuracy, interpretation, academic integrity, and final content of this submission.